

CV

Applied machine learning researcher working in healthcare and medical data science, with current emphasis on clinical microbiology, trustworthy biomedical AI, and open-source translational software. My long-term goal is to build an independent research group focused on clinical AI systems that help health professionals and biomedical researchers solve real problems with multimodal data.

My profile combines competitive funding, collaborative translational research, supervision, and open science. I hold a Marie Skłodowska-Curie Postdoctoral Fellowship (MSCA PF), and my work spans microbiology, speech, cardiology, and medical imaging under a common agenda of applied AI for health.

I have 24 papers (19 published or accepted in peer-reviewed journals and 5 under review), including 6 first/co-first-author and 2 senior/last-author papers. My [Google Scholar profile reports an h-index of 9](#) and 263 citations (updated August 31, 2026).

Recent acceptance: MARISMa - accepted in Scientific Data (2026); last author.

Academic background

- Ph.D. in Bayesian Machine Learning applied to Personalised Medicine, [Universidad Carlos III de Madrid](#), May 2023
- M.Sc. in Information Health Engineering, [Universidad Carlos III de Madrid](#), June 2020
- B.Sc. in Telematic Engineering, [Universitat de les Illes Balears](#), June 2019

Professional history

Academic experience

- 07/2025-: Postdoctoral researcher (Prof. Adrian Egli's group)
 - Company: [University of Zurich, Institute for Medical Microbiology, Applied Microbiology Research \(AMR\) lab](#)
 - Field: Applied ML to Microbiology
 - 07/2026-: Marie Skłodowska-Curie Postdoctoral Fellow (MSCA PF)
 - 07/2025-06/2026: Postdoctoral researcher
- 2024-2025: Visiting professor in Data science
 - Company: UC3M, Signal Theory and Communications Department
 - Field: Bayesian ML
- 2023-2024: Postdoctoral researcher
 - Company: [Universidad Politécnica de Madrid \(UPM\), Bioengineering and Optoelectronics Group](#)
 - Field: Bayesian ML applied to speech
 - Supervisor: [Prof. Juan Ignacio Godino Llorente](#)
 - Research output: First-author MARTA paper, A model for the automatic phonemic grouping of Parkinsonian speech, accepted in Engineering Applications of Artificial Intelligence (2026).
- 2020-2023: Ph.D. Candidate
 - Company: UC3M, Signal Theory and Communications Department and [Gregorio Marañón Health Research Institute, Microbiology and Infectious Diseases Department](#)
 - Field: Bayesian ML and its application to microbiology
 - Supervisors: [Prof. Pablo M. Olmos](#) and [Dr. Vanessa Gómez-Verdejo](#) at the university and [Dr. Belén Rodríguez-Sánchez](#) at the hospital.
- 2019-2020: Research Assistant
 - Company: Universidad Carlos III de Madrid (UC3M), Signal Theory and Communications Department
 - Field: Bayesian ML
 - Supervisor: [Prof. Pablo M. Olmos](#)
- 2017-2018: Research Assistant
 - Company: [Universitat de les Illes Balears, UGIVIA group](#)
 - Field: Computer Vision
 - Supervisors: [Prof. Francisco J. Perales](#) and [Dr. Silvia Ramis](#)

Industry experience

- 2018-2019: Telecommunication Engineer
 - Company: [ALCORT S.L.](#)
 - Field: Fiber to the Home
 - Supervisors: [Alejandro Cortés](#) and [Teresa Reus](#)

Research visits & collaborations

- 30 July-3 August 2026: Invited research visit to the Department of [Clinical Microbiology](#), Christian Medical College (CMC) Vellore, India. Project meetings and collaborative work on applications of artificial intelligence in clinical microbiology.

Grants obtained

- Marie Skłodowska-Curie Postdoctoral Fellowship (European Commission) 01/07/2026 - 30/06/2028
 - Project: OUTBRAID - OUTBreak AI-driven Detection: Enhancing MALDI-TOF outbreak detection with multimodal AI
 - Supervisor: Prof. Adrian Egli (University of Zurich)
 - Highly competitive: 17,066 applicants, 1,610 funded (9.6% success rate)

Publications

Peer-review journals

MALDI-TOF MS spectra database from a Spanish University General Hospital (2018-2024)

Accepted for publication in Scientific Data, 2026

MARISMa: an open MALDI-TOF MS resource. Last-author paper accepted for publication in Scientific Data; final publication details forthcoming.

Recommended citation: Santiago, L. S., López-Mareca, I., Blázquez-Sánchez, M., Moreno, J. M., Sevilla-Salcedo, C., Gómez-Verdejo, V., Rodríguez-Sánchez, B., Rodríguez-Temporal, D., & Guerrero-López, A. (2026). MALDI-TOF MS spectra database from a Spanish University General Hospital (2018-2024). Scientific Data. Accepted for publication.

A model for the automatic phonemic grouping of Parkinsonian speech

Accepted for publication in Engineering Applications of Artificial Intelligence, 2026

MARTA, developed during my postdoctoral research at UPM. First-author paper accepted in Engineering Applications of Artificial Intelligence; final publication details forthcoming.

Recommended citation: Guerrero-López, A., Arias-Londoño, J. D., Shattuck-Hufnagel, S., & Godino-Llorente, J. I. (2026). A model for the automatic phonemic grouping of Parkinsonian speech. Engineering Applications of Artificial Intelligence. Accepted for publication.

State-of-the-art in automated image-based species identification systems

Accepted for publication in Frontiers in Cellular and Infection Microbiology, 2026 · Review article

Review article (Mini Review) on AI-based analysis of culture-plate images for microbial identification and laboratory workflows. Accepted 7 September 2026; the final formatted article is forthcoming.

Recommended citation: Weber, L., Hussen, M. M., Veeraraghavan, B., Meijer, E., Gütlin, Y., Guerrero-López, A., Brüningk, S., Nolte, O., Jutzeler, C., & Egli, A. (2026). State-of-the-art in automated image-based species identification systems. Frontiers in Cellular and Infection Microbiology. Mini Review, accepted for publication. <https://doi.org/10.3389/fcimb.2026.1818023>

A systematic review of machine learning on clinical MALDI-TOF MS

Published in Briefings in Bioinformatics, 2026

Recommended citation: Schmidt-Santiago, L., Guerrero-López, A., Sevilla-Salcedo, C., Rodríguez-Temporal, D., Rodríguez-Sánchez, B., & Gómez-Verdejo, V. (2026). A systematic review of machine learning on clinical MALDI-TOF MS. Briefings in Bioinformatics, 27(3), bbag208. <https://doi.org/10.1093/bib/bbag208>

How ready are we to use artificial intelligence in our fight against antimicrobial resistance? An ESGAID and EAAS perspective

Published in Expert Review of Anti-infective Therapy, 2026 · Perspective

Published perspective from ESGAID and EAAS on AI for antimicrobial resistance. Last individual author.

Recommended citation: Giacobbe, D. R., Ahmad, R., Akilli, F. M., Ascandari, A., Eyre, D. W., Gallardo-Pizarro, A., ... & Guerrero-López, A., on behalf of ESGAID and EAAS. (2026). How ready are we to use artificial intelligence in our fight against antimicrobial resistance? An ESGAID and EAAS perspective. Expert Review of Anti-infective Therapy, 24(1), 39-61. <https://doi.org/10.1080/14787210.2026.2625382>

Transformer Vibration Forecasting for Advancing Rail Safety and Maintenance 4.0

Published in IEEE Open Journal of Intelligent Transportation Systems, 2025

Recommended citation: Larese, D. C., Cerrada, A. B., Tomei, G. D., Guerrero-López, A., Olmos, P. M., & Gómez García, M. J. (2025). Transformer Vibration Forecasting for Advancing Rail Safety and Maintenance 4.0. IEEE Open Journal of Intelligent Transportation Systems, 6, 1612-1624. <https://doi.org/10.1109/OJITS.2025.3638611>

ESCMID workshop: Artificial intelligence and machine learning in medical microbiology diagnostics

Published in Microbes and Infection, 2025

Recommended citation: Greutmann, M., Borgwardt, K., Brüningk, S., Franzeck, F., Giske, C. G., Green, A. G., ... & Egli, A. (2025). ESCMID workshop: Artificial Intelligence and Machine Learning in Medical Microbiology Diagnostics. Microbes and Infection, 105562. <https://doi.org/10.1016/j.micinf.2025.105562>

CT-SCOPE: annotated dataset of CT SCans for the automatic semantic segmentation of the Osseous structures of the Paranasal sinusEs

Published in Data in Brief, 2025

Recommended citation: Sun, Y., Guerrero-López, A., Arias-Londoño, J. D., & Godino-Llorente, J. I. (2025). CT-SCOPE: annotated dataset of CT SCans for the automatic semantic segmentation of the Osseous structures of the Paranasal sinusEs. Data in Brief, 111962. <https://doi.org/10.1016/j.dib.2025.111962>

Bayesian automatic screening of pneumonia and lung lesions localization from CT scans. A combined method toward a more user-centred and explainable approach

Published in IEEE Access, 2025

Recommended citation: Moure Prado, Á., Guerrero-López, A., Arias-Londoño, J. D., & Godino-Llorente, J. I. (2025). Bayesian automatic screening of pneumonia and lung lesions localization from CT scans. A combined method toward a more user-centred and explainable approach. IEEE Access, 13, 3607282. <https://doi.org/10.1109/ACCESS.2025.3607282>

Automated web-based typing of Clostridioides difficile ribotypes via MALDI-TOF MS

Published in BMC Bioinformatics, 2025

A web-based translational tool for automated Clostridioides difficile ribotype typing from MALDI-TOF MS data.

Recommended citation: Blázquez-Sánchez, M.*, Guerrero-López, A.*, Candela, A., Belenguer-Llorens, A., Moreno, J. M., Sevilla-Salcedo, C., ... & Rodríguez-Sánchez, B. (2025). Automated web-based typing of Clostridioides difficile ribotypes via MALDI-TOF MS. BMC Bioinformatics, 26(1), 181.
<https://doi.org/10.1186/s12859-025-06200-6>

Automatic semantic segmentation of the osseous structures of the paranasal sinuses

Published in Computerized Medical Imaging and Graphics, 2025

Medical image segmentation of paranasal sinus osseous structures for clinically relevant CT analysis.

Recommended citation: Sun, Y., Guerrero-López, A., Arias-Londoño, J. D., & Godino-Llorente, J. I. (2025). Automatic semantic segmentation of the osseous structures of the paranasal sinuses. Computerized Medical Imaging and Graphics, 123, 102541.
<https://doi.org/10.1016/j.compmedimag.2025.102541>

NeuroVoz: a Castilian Spanish corpus of parkinsonian speech

Published in Scientific Data, 2024

Recommended citation: Mendes-Laureano, J., Gómez-García, J. A., Guerrero-López, A., et al. (2024). NeuroVoz: a Castilian Spanish corpus of parkinsonian speech. Scientific Data, 11, 1367.
<https://doi.org/10.1038/s41597-024-04186-z>

A voice and speech corpus of patients who underwent upper airway surgery in pre- and post-operative states

Published in Scientific Data, 2024

Recommended citation: Hernández-García, E., Guerrero-López, A., Arias-Londoño, J. D., et al. (2024). A voice and speech corpus of patients who underwent upper airway surgery in pre- and post-operative states. Scientific Data, 11, 746. <https://doi.org/10.1038/s41597-024-03540-5>

Exploring the Power of Photoplethysmogram Matrix for Atrial Fibrillation Detection with Integrated Explainability

Published in Engineering Applications of Artificial Intelligence, 2024

Explainable deep learning for atrial fibrillation detection from PPG-derived matrix representations.

Recommended citation: Fuster-Barceló, C., Guerrero-López, A., Camara, C., & Peris-Lopez, P. (2024). Exploring the Power of PPG Matrix for Atrial Fibrillation Detection with Integrated Explainability. Engineering Applications of Artificial Intelligence, 133, 108325.
<https://doi.org/10.1016/j.engappai.2024.108325>

Detecting train driveshaft damages using accelerometer signals and Differential Convolutional Neural Networks

Published in Engineering Applications of Artificial Intelligence, 2023

Recommended citation: López, A., Guerrero-López, A., Olmos, P. M., & Gómez, M. J. (2023). Detecting train driveshaft damages using accelerometer signals and Differential Convolutional Neural Networks. Engineering Applications of Artificial Intelligence, 126, 106840.
<https://doi.org/10.1016/j.engappai.2023.106840>

Automatic Discrimination of Species within the Enterobacter cloacae Complex Using Matrix-Assisted Laser Desorption Ionization-Time of Flight Mass Spectrometry and Supervised Algorithms

Published in Journal of Clinical Microbiology, 2023

Supervised learning for species-level discrimination within the Enterobacter cloacae complex using routine MALDI-TOF measurements.

Recommended citation: Candela, A.*, Guerrero-López, A.*, Mateos, M., Gómez-Asenjo, A., Arroyo, M. J., Hernández-García, M., del Campo, R., Cercenado, E., Cuénod, A., Méndez, G., Mancera, L., Caballero, J. D., Martínez-García, L., Gijón, D., Morosini, M. I., Ruiz-Garbajosa, P., Egli, A., Cantón, R., Muñoz, P., Rodríguez-Temporal, D., & Rodríguez-Sánchez, B. (2023). Automatic Discrimination of Species within the Enterobacter cloacae Complex Using Matrix-Assisted Laser Desorption Ionization-Time of Flight Mass Spectrometry and Supervised Algorithms. Journal of Clinical Microbiology, e01049-22. <https://journals.asm.org/doi/abs/10.1128/jcm.01049-22>

Automatic antibiotic resistance prediction in Klebsiella pneumoniae based on MALDI-TOF mass spectra

Published in Engineering Applications of Artificial Intelligence, 2023

Machine learning for rapid antibiotic-resistance prediction in Klebsiella pneumoniae directly from routine MALDI-TOF mass spectra.

Recommended citation: Guerrero-López, A., Sevilla-Salcedo, C., Candela, A., Hernández-García, M., Cercenado, E., Olmos, P. M., Cantón, R., Muñoz, P., Gómez-Verdejo, V., del Campo, R., & Rodríguez-Sánchez, B. (2023). Automatic antibiotic resistance prediction in Klebsiella pneumoniae based on MALDI-TOF mass spectra. Engineering Applications of Artificial Intelligence, 118, 105644. <https://www.sciencedirect.com/science/article/pii/S0952197622006340>

Bayesian Sparse Factor Analysis with Kernelized Observations

Published in Neurocomputing, 2022

Recommended citation: Sevilla-Salcedo, C., Guerrero-López, A., Olmos, P. M., & Gómez-Verdejo, V. (2022). Bayesian Sparse Factor Analysis with Kernelized Observations. Neurocomputing. <https://doi.org/10.1016/j.neucom.2022.03.024>

Evaluation of a VR system for Pain Management using binaural acoustic stimulation

Published in Multimedia Tools and Applications, 78, 2019

Recommended citation: Perales, F. J., Riera, L., Ramis, S., Guerrero-López, A. (2019). Evaluation of a VR system for Pain Management using binaural acoustic stimulation. Multimed Tools Appl, 78, 32869-32890. <https://link.springer.com/article/10.1007/s11042-019-07953-y>

Preprints

MALDI-ST: A deep learning-based framework for rapid bacterial strain typing using MALDI-TOF mass spectra

Published in medRxiv, 2026

A deep learning framework for rapid MALDI-TOF-based strain typing across four clinically important bacterial species, including external validation and interpretable spectral signatures.

Recommended citation: Nguyen, H.-A., Peleg, A. Y., Song, J., Vezina, B., Egli, A., Guerrero-López, A., Blakeway, L. V., Wisniewski, J. A., Badoordeen, G. Z., Theegala, R., Doan, N. Q., Dowe, D. L., & Macesic, N. (2026). MALDI-ST: A deep learning-based framework for rapid bacterial strain typing using MALDI-TOF mass spectra. medRxiv. <https://doi.org/10.64898/2026.08.08.26359928>

Development and validation of a LAMP assay for rapid detection of Escherichia coli associated UTIs with increased risk of urosepsis

Published in Research Square, 2026

A rapid LAMP-based assay for detecting Escherichia coli and the urosepsis-associated papGII virulence factor directly from urine specimens.

Recommended citation: Rogenmoser, J., Guerrero-López, A., Nolte, O., Quiblier, C., Hinic, V., Kolesnik-Goldmann, N., Biggel, M., & Egli, A. (2026). Development and validation of a LAMP assay for rapid detection of Escherichia coli associated UTIs with increased risk of urosepsis. Research Square. <https://doi.org/10.21203/rs.3.rs-9836245/v1>

Automatic surveillance of Escherichia coli bacteriological strains within clinical settings

Published in bioRxiv, 2024

A machine learning approach for automated strain surveillance of Escherichia coli in clinical settings.

Recommended citation: Rodríguez Palomo, R., Guerrero-López, A., Sevilla-Salcedo, C., Rodríguez-Temporal, D., Rodríguez-Sánchez, B., & Gómez-Verdejo, V. (2024). Automatic surveillance of Escherichia coli bacteriological strains within clinical settings. bioRxiv. <https://doi.org/10.1101/2024.11.05.622049>

Overcoming variability challenges for Clostridioides difficile via data augmentation techniques

Published in bioRxiv, 2024

Recommended citation: Guerrero-López, A., Antón, L. B., Blázquez-Sánchez, M., Schmidt-Santiago, L., Sevilla-Salcedo, C., Rodríguez-Temporal, D., Rodríguez-Sánchez, B., & Gómez-Verdejo, V. (2024). Overcoming variability challenges for Clostridioides difficile via data augmentation techniques. bioRxiv. <https://doi.org/10.1101/2024.10.29.620907>

Multi-view hierarchical Variational AutoEncoders with Factor Analysis latent space

Published in arXiv, 2022

Recommended citation: Guerrero-López, A., Sevilla-Salcedo, C., Gómez-Verdejo, V., & Olmos, P. M. (2022). Multi-view hierarchical Variational AutoEncoders with Factor Analysis latent space. arXiv. <https://arxiv.org/abs/2207.09185>

Conference proceedings

Interacción basada en robots sociales para la evaluación de expresiones faciales

Published in Interaccion 2019 at San Sebastián, Spain., 2019

Recommended citation: Ramis, S., Perales, F., Guerrero-López, A., & Buades Rubio, J. M. (2019). Interacción basada en robots sociales para la evaluación de expresiones faciales. https://www.researchgate.net/publication/334204484_Interaccion_basada_en_robots_sociales_para_la_evaluacion_de_expresiones_faciales

Datasets

MARISMa: a routine MALDI-TOF MS database from 2018 to 2024

Published in Zenodo, 2025

Recommended citation: Santiago, L. S., López-Mareca, I., Blázquez-Sánchez, M., Moreno, J. M., Sevilla-Salcedo, C., Gómez-Verdejo, V., Rodríguez-Sánchez, B., Rodríguez-Temporal, D., & Guerrero-López, A. (2025). MARISMa: a routine MALDI-TOF MS database from 2018 to 2024. Zenodo. <https://zenodo.org/records/14218003>

A voice and speech corpus of patients who underwent upper airway surgery in pre- and post-operative states

Published in Zenodo, 2024

Recommended citation: Hernández-García, E., Guerrero-López, A., Arias-Londoño, J. D., et al. (2024). CUCO Database: A voice and speech corpus of patients who underwent upper airway surgery in pre- and post-operative states [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.10256802>

NeuroVoz: a Castilian Spanish corpus of parkinsonian speech

Published in Zenodo, 2024

An open Castilian Spanish speech corpus designed to support Parkinson-related speech analysis and digital biomarker research.

Recommended citation: Mendes-Laureano, J., Gómez-García, J. A., Guerrero-López, A., Luque-Buzo, E., Arias-Londoño, J. D., Grandas-Pérez, F. J., & Godino-Llorente, J. I. (2024). NeuroVoz: a Castilian Spanish corpus of parkinsonian speech (1.0.0) [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.10777657>

MALDI-TOF spectra of Clostridioides difficile Ribotypes

Published in Zenodo, 2023

Recommended citation: Blázquez-Sánchez, M., Guerrero-López, A., Candela, A., Sánchez-Cueto, M., Marín, M., Jiménez-Navarro, L., et al. (2023). MALDI-TOF spectra of Clostridioides difficile Ribotypes. Zenodo. <https://doi.org/10.5281/zenodo.10370872>

Misc

Clinical Microbiology with Multi-view deep probabilistic models

Published in Universidad Carlos III de Madrid, 2023

Recommended citation: Guerrero-López, A. J. (2023). Clinical microbiology with multi-view deep probabilistic models. http://aguerrerolopez.github.io/files/Tesis_GuerreroLopez_AlejandroJorge_FormatoWeb.pdf

Teaching

I have contributed to 11 taught courses, supervised 3 BSc theses and 8 MSc theses, and I currently co-supervise 4 PhD students.

Ongoing PhD co-supervision (in progress)

- Luis Fuertes Pina - Development of machine learning models applied to FTIR and MALDI-TOF for improving microbiological diagnosis. Universidad Complutense de Madrid (UCM). Supervisors: Rafael María Cantón Moreno, José Avendaño Ortiz, and Alejandro Guerrero-López. Ongoing.
- Janis Rogenmonser - Advancing antimicrobial resistance prediction from whole-genome sequencing. Main supervisor: Prof. Adrian Egli. Co-supervisor: Alejandro Guerrero-López. Started September 2025; co-supervising since September 2025.
- Eline Meijer - Boosting the resolution of MALDI-TOF mass spectra using transformer-based machine learning. Main supervisor: Prof. Adrian Egli. Co-supervisor: Alejandro Guerrero-López. Started January 2025; co-supervising since July 2025.
- Yukino Gütlin - Predicting antimicrobial resistance, virulence, and invasiveness from MALDI-TOF MS data. Main supervisor: Prof. Adrian Egli. Co-supervisor: Alejandro Guerrero-López. Started 2024; co-supervising since July 2025.

Undergraduate Courses

- [BIO296 - Microbial bioinformatics: sequencing technologies to pathogen analysis - University of Zurich](#), BSc Biology block course (6 ECTS) (2026)
- [Neural Networks](#) - Universidad Carlos III de Madrid, Bachelor in Data Science and Engineering (2025)
- [Machine Learning applications](#) - Universidad Carlos III de Madrid, Bachelor in Data Science and

Engineering (2025)

- [Linear networks analysis and synthesis](#) - Universidad Carlos III de Madrid, Bachelor in Mobile and Space Communications Engineering (2025)
- [Neural Networks](#) - Carlos III International School, Bachelor in Data Science and Engineering (2022)
- [Machine Learning Applications](#) - Carlos III International School, Bachelor in Data Science and Engineering (2022)
- [Modern Theory of Detection and Estimation](#) - Universidad Carlos III de Madrid, Signal Theory and Communications Department (2021)

Bachelor Theses

- [Automated measurement of disc diffusion images using object detection](#) - Bachelorstudium Medizininformatik, ZHAW School of Engineering (2026) Completed. Defended 01 July 2026. Grade: 4.5/6.
- [Connecting MALDI-TOF MS and whole-genome sequencing data in routine diagnostics](#) - Bachelorstudium Medizininformatik, ZHAW School of Engineering (2026) Completed. Defended 01 July 2026. Grade: 5.5/6.
- [Characterizing train driveshaft damages using accelerometers](#) - Bachelor in Data Science and Engineering, Universidad Carlos III de Madrid (2023)

Master Programs

- [Bio-inspired Learning](#) - Universidad Politécnica de Madrid, Master in Theoretical Signal Processing (2024)
- [AI in Health](#) - Universidad Carlos III de Madrid, Master in Applied Artificial Intelligence (2023)
- [Deep Learning](#) - Universidad Carlos III de Madrid, Master in Applied Artificial Intelligence (2022)

Master Theses

- [Automatic unsupervised domain adaptation of MALDI-TOF MS](#) - Master in Applied Artificial Intelligence, Universidad Carlos III de Madrid (2025)
- [Fungi automatic classifier using machine learning methods](#) - Master in Applied Artificial Intelligence, Universidad Carlos III de Madrid (2025)
- [Conditional Diffusion models for audio generation: from EEG to Speech](#) - Master in Applied Artificial Intelligence, Universidad Carlos III de Madrid (2024)
- [Adversarial Training in Microbiology: testing robustness](#) - Master in Applied Artificial Intelligence, Universidad Carlos III de Madrid (2024)
- [Antibiotic recommender system based on MALDI-TOF and Neural Networks](#) - Master in Applied Artificial Intelligence, Universidad Carlos III de Madrid (2024)
- [Clustering of bacteriological strains using unsupervised learning and mass spectrometry](#) - Master in Applied Artificial Intelligence, Universidad Carlos III de Madrid (2023)
- [Predicting antibiotic-resistant bacteria using mass spectrometry and deep learning](#) - Master in Applied Artificial Intelligence, Universidad Carlos III de Madrid (2023)
- [A machine learning approach to League of Legends pre-game winning team prediction](#) - Master in Big Data Analytics, Universidad Carlos III de Madrid (2023)

Corporate Training

- [Machine Learning Fundamentals](#) - Fundación Carlos III de Madrid (2022)

Teaching innovation projects

Co-coordinator with Alejandro Lanco Serrano of UC3M teaching innovation project No. 25, academic year 2024-2025: Combinación de conocimiento en redes neuronales e IA generativa para participar en competición real de ingeniería de datos (combining neural networks and generative AI knowledge to participate in a real data engineering competition).

Supported by Universidad Carlos III de Madrid under its 22nd Call for Teaching Innovation Experiences. Neural Networks course, Department of Signal Theory and Communications. Degree programmes: BSc in Data Science and Engineering and the Double Degree in Data Science and Engineering and Telecommunication Technologies Engineering. Two participating lecturers; one-semester duration.

Teaching evaluations & Recognitions

- Based on 9 official teaching evaluations, achieved an average score of 4.83 / 5 with a standard deviation of 0.29.
- Recipient of multiple formal letters of recognition from Universidad Carlos III de Madrid for

outstanding teaching performance.
- Full record of evaluations and recognition letters (PDF)

Invited Talks

2026

AI in diagnostics, MALDI-TOF MS, and image recognition

August 05, 2026

Invited talk at BioMET '26 - International Conference on Biomaterials, Bioengineering, and Biotheranostics, Vellore Institute of Technology (VIT), Vellore, Tamil Nadu, India

AI applications in clinical microbiology

July 30-August 3, 2026

Invited talk at Department of Clinical Microbiology, Christian Medical College (CMC) Vellore, Vellore, Tamil Nadu, India

What does an AI scientist actually do in a microbiology institute?

2026

Invited talk at Master program Applied Computational Life Science, ZHAW, ZHAW, Switzerland

Talks

2026

Democratising MALDI-TOF AMR prediction: from Transformers to simple Neural Networks without accuracy loss, reducing energy costs via knowledge distillation

April 21, 2026

Oral abstract presentation at ESCMID Global 2026 - 36th Congress of the European Society of Clinical Microbiology and Infectious Diseases, Munich, Germany

Digital AST Annotator: standardising Kirby-Bauer reading with an open-source app

April 20, 2026

Oral abstract and ePoster presentation at ESCMID Global 2026 - 36th Congress of the European Society of Clinical Microbiology and Infectious Diseases, Munich, Germany

2023

Clinical microbiology with multi-view deep probabilistic models

May 31, 2023

PhD defense at PhD Defense, Universidad Carlos III de Madrid, Universidad Carlos III de Madrid, Madrid, Spain

2022

Multi-view hierarchical Variational AutoEncoders with Factor Analysis latent space

July 16, 2022

Seminar presentation at Cambridge Ellis Machine Learning Summer School, University of Cambridge, Cambridge, United Kingdom

Identification of subspecies within Enterobacter Cloacae Complex by MALDI-TOF MS and Machine Learning

June 08, 2022

Seminar presentation at 9ª Jornada de Investigación e Innovación IISGM, Hospital General Universitario Gregorio Marañón, Madrid, Spain

Application of Machine Learning for the prediction of antimicrobial resistance mechanisms: a semi-supervised heterogeneous approach

March 07, 2022

Seminar presentation at Servicio de Microbiología, Hospital General Universitario Gregorio Marañón, Madrid, Spain

Application of Machine Learning for the prediction of antimicrobial resistance mechanisms: a semi-supervised heterogeneous approach

January 26, 2022

Seminar presentation at Seminarios Alberto Tejedor, Hospital General Universitario Gregorio Marañón, Madrid, Spain

2021

MSACL 2021 EU 7th European Congress

September 13, 2021

Podium presentation at Mass Spectrometry & Advances in the Clinical LAB 2021, Salzburg, Austria

Reviewing

- IEEE JBHI, [certificate](#), [IEEE JBHI, 2024](#) - currently
- mSystems, [certificate](#), [mSystems](#), [American Society for Microbiology, 2024](#)
- Engineering Applications of Artificial Intelligence, [certificate](#), [EAAI, 2023](#) - currently
- Artificial Intelligence and Statistics, [certificate](#), [AISTATS, 2023](#) - currently
- Clinical Microbiology and Infection, CMI, [2023](#) - currently

Training & Professional Development

- Grant writing workshop for postdocs, University of Zurich Graduate Campus, 04/02/2026-13/02/2026 (Instructor: PhD Ingo Hebach)
- Project management for postdocs, University of Zurich Graduate Campus, 18/09/2025-19/09/2025 (Instructor: Dr. Carine Galli Marxer)
- [Cambridge Ellis Machine Learning Summer School](#), University of Cambridge, Cambridge, United Kingdom, 2022
- Digital Health, the regulatory landscape, Medidee, [Laussane](#), Switzerland, 2022
- Teaching trainee at DEMOTEC, UIB, Palma, Spain, 2016
- Volunteer instructor at the Science Fair at the UIB , 2015 [and](#) 2016

Workshops

Predicting negative blood cultures from EHR using Orange Data Mining

August 04, 2026

Hands-on workshop at BioMET '26 pre-conference workshop: Clinical AI Transforming Diagnosis, Treatment & Beyond, Vellore Institute of Technology (VIT), Vellore, Tamil Nadu, India

ESCMID Workshop on Artificial Intelligence and Machine Learning in Medical Microbiology Diagnostics

June 2-5, 2025

Instructor at ESCMID course organised by ESGMD and the Swiss Society for Microbiology, University of Zurich, Zurich, Switzerland

2nd Automatic Assessment of Parkinsonian Speech Workshop (AAPS'24)

November 21-22, 2024

Local organising committee member at Research Laboratory of Electronics (RLE), Massachusetts Institute of Technology (MIT), Cambridge, Massachusetts, USA